Final Exam 2022 Subjectives of Zoo103
By Muhammad Faryad

Q:-Define Herbivores with example?
Herbivores-**Animals** that only eat plant. … Rabbits, Goat, sheep, grasshoppers etc
Q:-Discribe Jassids

1. Nymphs and adults both suck sap.
2. Leaves first show yellow and then reddish spots.
3. Attacked leaves turn downwards, dry and fall
4. Plants become weak and their fruit fall

Q:-Identification of jumping spider? 2
Jumping spiders are generally recognized by their eye pattern.
All jumping spiders have four pairs of eyes, with the anterior median pair being particularly large
Q:-Define parasitoid 2
A **parasitoid** is an organism that has young that develop on or within another organism (the host), eventually killing it.

Parasitoids have characteristics of both predators and parasites.

Parasitoids are usually smaller than their selected host
Q:-What are the classifications of Arthropoda?
Chelicerata (arachnids), Crustacea (crustaceans), Hexapoda (insects and springtails), and Myriapoda (millipedes and centipedes).
Q:-Examples of Annelids 2

1. Earthworm
2. Leeches
3. Polychaetes

Q:-Define Phylogram 2
Phylogram is a tree withbranching that are proportional to the evolutionary distances.
Q:-Morphology of Scorpions or Body Parts of Scorpions 2

1. Cephalothorax
2. Padipal
3. Chelicera
4. Abdomen
5. Walkinglegs
6. Sting
7. Body is covered with a cuticle consisting of chitin and tanned proteins

Q:-<u>Define acculmulation curve 2</u>
The species discovery curve or species accumulation curve is a graph recording the cumulative number of species of living things recorded in a particular environment as a function of the cumulative effort expended searching for them.
Q:-<u>Define polytype species 2</u>
A species contains two or more subspecies• Examples are tiger, *Panthera tigris* which has several subspecies.

Q:-Two way Anova 2

In two-way ANOVA there are two independent variables and one dependent variable.

Q:-Define monogram 2

A sign of identify usually formed ofthe combined initial name .

Q:-Epigyne 2

Females genetal organ is also present on ventral side of the abdomen. It is called epigyne.

Q:-Peripatric speciation 2

Peripatric speciation is a mode of speciation in which a new species is formed from an isolated peripheral population.

Q:-Punctuated gradualism 2

In evolutionary biology, punctuated equilibrium (also called punctuated equilibria) is a theory that proposes that once a species appears in the fossil record, the population will become stable, showing little evolutionary change for most of its geological history.

In one-way ANOVA there was only one independent variable .The population must be close to a normal distribution. Samples must be independent.

Population variances must be equal.

Q:-Four lineages of arthropods 2

1. **Trilobites** (extinct
2. **Chelicerates** (horseshoe crabs, spiders
3. **Uniramians** (centipedes, millipedes, insects
4. **Crustaceans**)crabs, lobsters, barnacles

Q:-Who introduced pronghorn 2

In 1855 John Edward Gray published the name of Pronghorn.

Q:-Define standard deviation 2

The standard deviation is the square root of the variance.

$$SD = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

Q:-Difference b/w Sample standard deviation and population standard deviation 3

• A sample standard deviation is a statistic. This means that it is calculated from only some of the individuals in a population.

• The population standard deviation is a parameter, which is a fixed value calculated from every individual in the population.

Q:-classification of green lacewings 3

Phylum	Arthropoda
Class	Insecta
Family	Chrysopidae
Genus	Chrysoperla
Species	C. carnea

Q:-Classification of whitefly.

TAXONOMY

Phylum	Arthropoda
Class	Insecta
Order	Hemiptra
Family	Aleyrodidae

Q:-Classification of Bug

Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Family	Pentatomidae
Sub-Family	Asopinae

Q:-Classification of fly upto family 3

Phylum	>Arthropoda ,
Class	>insecta ,
order	>dipetra
Family	

Q:-Classification of Bollworm 3

Phylum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Noctuoidea
Genus	Helicoverpa
Species	*H. armigera*

Q:-how many life stage of bollworm a are in one year

There are **four** distinct stages in the life cycle of the bollworm, as is the case with all butterflies and moths—namely, the egg, larva, pupa, and adult.

Q:-Paracitised 3

A parasitoid is an organism that has young that develop on or within another organism (the host), eventually killing it. • **Parasitoids** have characteristics of both predators and parasites.

Q:-Allelic frequiences of(p2 q2 2pq) 3

P = frequency of Allele A

q = Frequency of Allele a

2pq= Frequency of Aa genotype

Q:-What is a homologous structure? 3

A body part with the same structure between some groups due to inheritance from the same ancestor.
An example of homologous structures are the limbs of humans, cats, whales, and bats

Q:-Conditions for t test 3

The conditions required to conduct a t-test include the measured values in ratio scale or interval scale, simple random extraction, homogeneity of variance, appropriate sample size, and normal distribution of data.

Q:- How silk gland secreted 3

The silk gland is the only organ where silk proteins are synthesized and secreted in the silkworm, Bombyx mori. Silk proteins are stored in the lumen of the silk gland for around eight days during the fifth instar. Determining their dynamic changes is helpful for clarifying the secretion mechanism of silk proteins.

Q:-Amniotes Vs Anamniotes 3

Amniotes are higher vertebrates including reptiles, birds, and mammals
Anamniotes are lower vertebrates including fishes and amphibians

Q:-Mullerian Mimicry 3

Mullerian mimicry in the form of protective mimicry in which two or more distasteful or harmful species especially insects, closely resemble each other and therefore avoided equally by all their natural predators

Q:-Parametric test list 3

Parametric tests are used for
Normal data •,T-test •, Z-test •, F-test• ,ANOVA ,Regression ,correlation

Q:-Difference between Priori & Posteriori 3

Priori

(before/without phyletic information) • Proposed by Artistotle

Posteriori

• Proposed by Adenson (after phyletic information collection) • All classification is based on this model.

Q:-Three types of sponges 3

1. , the Hexactinellida (glass sponges),
2. the Demospongia, and
3. the Calcariea (calcareous sponges).

Q:-Taxonomy and Ecology of Yellow stem borer Control 5

1. Destruction of weeds harbouringpests
2. Pulling out infested rice plant
3. Flooding the field after harvest
4. Stubble (short cut stalks) leftafter harvest should be
5. ploughed up and burn orsubmerged in water immediately after harvest
6. Crop rotation

Q:-What are the functions of the mammalian glands? 5

Functionally, the mammary glands **produce milk**; structurally, they are modified sweat glands. Mammary glands, which are located in the breast overlying the pectoralis major muscles, are present in both sexes, but usually are functional only in the female.

Q:-Cephalothorax of spider 5

Cephalic region has 6-8 eyes •Eyes or black or diurnal •White or nocturnal •If only one type of eyes (homogenous) •If both type of eyes or present(heterogeneous) Usually arranged in two rows•Each row usually contain 4 eyes. •Rows of eyes are frequently curved. •Anterior median/Anterior lateral •Posterior median/Posterior lateral• Cephalic area of eyes is called **ocular area.** •Area of four median eyes is called ocular quad. •The area between anterior row of eyes and anterior margin of carapace is **clypeus** •The depression in the middle of thorax

is called **thoracic groove or fovea.**

Q:-on the basis of coelom animal are divided into how many groups and example also 5
Coelom
Body cavity between the body wall and alimentary canal

1. **Acoelomates** Coelenterates, Platyhelminthes
2. **Coelomates** Mollusk, Annelids, Arthropod, Echinoderms, Vertebrates etc
3. **Pseudocoelomates** Nematodes

Q:-Charecterstics of chordates 5
Five characteristic featuresof chordates present during some time of their life cycles are

1. a notochord,
2. a dorsal hollow tubular nerve cord,
3. pharyngeal slits,
4. endostyle/thyroid gland,
5. and a post-anal tail

Q:-characteristic of Reptiles 5

1. Reptiles are the creeping and burrowing cold-blooded
2. vertebrates bearing epidermal scales.
3. They are ectothermic (cold-blooded) and are found mostly in the warmer parts of the world.
4. Skin is dry, rough and without glands bearing epidermal scales or scutes.
5. Snakes and lizards shed their scales as skin cast.
6. They do not respire by means of gills.Respiration always takes place through lungs.
7. Skull is monocondylic,i.e.,with single occipital condyle.
8. Except in snakes, there are two pairs of pentadactyl limbs, each with 5 digits bearing claws—tetrapodus pentadactyl type.

Q:-Composition of silkworm. 5
Silkworm consists of two main proteins, sericin and fibroin, fibroin being the structural center of the silk, and serecin being the sticky material surrounding it.
Chemical Composition of Silk Fiber: · Fibroin: 75% · Ash of Silk Fibroin: 0.5% · Sericin: 22.5% · Fat and Wax: 1.5% · Mineral Salt: 0.5%

Q:-The important characteristics of arthropoda include:5

1. They possess an exoskeleton.
2. They have jointed appendages.
3. Their body is segmented.
4. They are bilaterally symmetrical.
5. They possess an open circulatory system.

Q:-Note on sericulture in Pakistan 5

1. Despite the growing demand for natural silk, this industry could not flourish in Pakistan.
2. This industry is facing numerous challenges such as lack of trained personnel, high mortality rate in local silk breed, and poor economic conditions of the farmers. Government.
3. Provide livelihood for rural poor families but silk proteins also have wide range of applications in textile, food, cosmetics and pharmaceutical industry as well as in traditional medicines.
4. Currently, Pakistan is importing about 70% raw silk to meet the demand of local market.
5. Unfortunately, this industry has not been on the list of priorities of our

Q:-General characteristics of Phylum Coelenterata 5

1. Coelenterata is also known as Cnidaria.
2. This name of cnidaria is due to the single cavity Coelenteron.
3. Radial Symmetry
4. Diploblastic
5. No separate body cavity.
6. They have an incomplete digestive systems. It has a single aperture mouth but doesn't have an anus.
7. They respire through the body surface by diffusion.
8. They excrete through body surface by diffusion.
9. Segmentations are absent in them.
10. They are mostly marine, aquatic animals and free living animals.

Q:-Mollusca Characteristics 5

1. They are mostly found in marine and freshwater. ...
2. They exhibit organ system level of organization.
3. Their body has a cavity.
4. The body is divided into head, visceral mass, muscular foot and mantle.
5. The head comprises of tentacles and compound eyes.
6. The body is covered by a calcareous shell.
7. They show bilateral symmetry.
8. They have a complete digestivetract.
9. They have open blood vascular system.

Q:-<u>Rice leaf folder mode of demage and control 5</u>

Damage

1. Larval stage is damaging
2. All stages of plants are damaged by this pest
3. Newlyhatched caterpillar cut leaf edges and folds the leaf
4. Infested leaves appear white
5. Attacked plant dries up
6. Yield loss vary from 10-50%

Control

1. Reform tillage and cultivation system
2. Light trap or lure can be used.
3. Utilize natural enemies (beetles and *Trichogramma* spp.) as biological control
4. Application of chlorpyrifos EC 1000ml/acre

Q:-characteristic of spiders 5

1. These characteristics include bodies divided into two tagmata (sections or segments),
2. eight jointed legs,
3. no wings or antennae,
4. the presence of chelicerae and pedipalps,
5. simple eyes, exoskeleton, which is periodically shed.

Q:-Hardy Weinberg Equilibrium Model 5

In a population, there are 320 AA individuals, 160 heterozygous and 20 aa individuals Calculate the genotypic andallelic frequencies,

Genotypes	AA	Aa	aa
Number of individuals	320	160	20
Genotypic frequencies	320/500=0.64	16/500=0.32	20/500=0.04

Allelic Frequencies	
A	320+320+160/1000 =0.8
a	160+40/1000 =0.2